# 3.32 Dedekind Domains

We develop the theory of Dedekind Domains. These occur both as number fields and regular affine curves (over any field). The important results are the finiteness of integral closure in a finite extension, and the ramification formula of a prime ideal. In the case of regular curves over a finite field further results are needed, which we prove later in the context of affine algebras.

**Proposition 3.32.1**
*Let $A$ be a Noetherian integral domain of dimension 1. Then the following are equivalent*

*a) $A$ is integrally closed*
*b) $A_\mathfrak{p}$ is integrally closed for every prime ideal $\mathfrak{p}$*
*c) $A_\mathfrak{p}$ is a discrete valuation ring for every non-zero prime ideal $\mathfrak{p}$*

*In this case we say that $A$ is a **Dedekind Domain**.*

*Proof.* Let $\mathfrak{p}$ be non-zero prime ideal of $A$. Then firstly $A_\mathfrak{p}$ is Noetherian by (...) and an integral domain by (...). Then $\dim A_\mathfrak{p} = \operatorname{ht}(\mathfrak{p}) \leq \dim A = 1$. However $(0) \subsetneq \mathfrak{p}$ so $\dim A_\mathfrak{p} = \operatorname{ht}(\mathfrak{p}) = 1$.

a) $\iff$ b) This is (3.24.10).

b) $\implies$ c) We have shown that $A_\mathfrak{p}$ is a Noetherian integral domain of dimension 1. Then it is a discrete valuation ring by (3.31.2).

c) $\implies$ b) Also follows from (3.31.2), since $A_{(0)}$ is a field and therefore automatically integrally closed. ∎

**Corollary 3.32.2**
*Every principal ideal domain which is not a field is a Dedekind Domain. For example $k[X]$ and $\mathbb{Z}$.*

*Proof.* This follows from (3.17.23) (3.24.12) (3.16.2). ∎

The following is useful in the theory of algebraic curves and function fields.

**Proposition 3.32.3 (Correspondence between Prime Ideals and Valuation Rings)**
*Let $A$ be an integral domain with Krull dimension 1 and quotient field $K$. Then there is a bijection*

$$\{\mathfrak{p} \lhd A \mid \mathfrak{p} \text{ non-zero prime}\} \longleftrightarrow \{A \subsetneq R \subsetneq K \text{ local rings}\}$$
$$\mathfrak{p} \longrightarrow A_\mathfrak{p}$$
$$\mathfrak{m}_R \cap A \longleftarrow R$$

*When $A$ is a Dedekind domain then all such local rings (and valuation rings) are discrete valuation rings.*

*Proof.* Suppose that $x \in \mathfrak{p}_1 \setminus \mathfrak{p}_2$. Then $x^{-1} \in A_{\mathfrak{p}_2}$ but we claim $x^{-1} \notin A_{\mathfrak{p}_1}$. For suppose $x^{-1} \in A_{\mathfrak{p}_1}$ then $\frac{ax}{b} = 1$ for $a \in A$ and $b \notin \mathfrak{p}_1$ whence $ax = b$ a contradiction. Therefore the given map is injective.

Let $R$ be a local ring with maximal ideal $\mathfrak{m}_R$. Then $\mathfrak{m}_R \cap A$ is a prime ideal, and so either $(0)$ or a non-zero prime $\mathfrak{p}$. We have $0 \neq \frac{x}{y} \in \mathfrak{m}_R$ for some $x, y \in A$ whence $0 \neq x \in \mathfrak{m}_R$. Therefore $\mathfrak{m}_R \cap A = \mathfrak{p}$. Suppose $x \in A \setminus \mathfrak{p}$ then $x \in R \setminus \mathfrak{m}_R \implies x \in R^\star$. This shows that $A_\mathfrak{p} \subset R$ and $\mathfrak{p}A_\mathfrak{p} \subset \mathfrak{m}_R$, and we are done by (3.25.13). ∎

**Proposition 3.32.4**
*Let $A$ be a Dedekind domain. Then every fractional ideal is invertible.*

*Proof.* Let $I$ be an invertible ideal. By (3.30.10) it is sufficient to show that $I_\mathfrak{p}$ is an invertible fractional ideal of $A_\mathfrak{p}$ for all prime ideals $\mathfrak{p}$. By (3.32.1) $A_\mathfrak{p}$ is a discrete valuation ring and the result follows from (3.31.7). ∎

**Lemma 3.32.5**
*Let $A$ be a Dedekind domain and $\mathfrak{a}$ a proper ideal. Then there is a well-defined group homomorphism*

$$(\mathfrak{a})^- : \mathbb{Z} \longrightarrow \mathcal{I}_A$$

*mapping 1 to $\mathfrak{a}$.*

*Proof.* By (3.32.4) the monoid $\mathcal{I}_A$ is a group, and therefore the first homomorphism exists by (3.2.6). ∎

**Lemma 3.32.6**
*Let $A$ be a Dedekind domain and $\mathfrak{p}$, $\mathfrak{q}$ non-zero prime ideals. Then*

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$$\left(\mathfrak{p}^k\right)_\mathfrak{q} = \begin{cases} A_\mathfrak{q} & \mathfrak{p} \neq \mathfrak{q} \\ \left(\mathfrak{q}A_\mathfrak{q}\right)^k & \mathfrak{p} = \mathfrak{q} \end{cases}$$

*Proof.* When $k = 0$ then by definition $\mathfrak{p}^k = A$ and $\left(\mathfrak{q}A_\mathfrak{q}\right)^k = A_\mathfrak{q}$ and so we are done.

We suppose that $k = 1$. If $\mathfrak{p} \neq \mathfrak{q}$ then as $\dim A = 1$ we have $\mathfrak{p} \not\subseteq \mathfrak{q}$. Therefore by definition $\mathfrak{p}_\mathfrak{q}$ contains 1 and equals $A_\mathfrak{q}$ as required. The case $\mathfrak{p} = \mathfrak{q}$ follows from (3.30.7).

By (3.30.8) and (3.32.5) both left and right hand sides define group homomorphisms $\mathbb{Z} \to \mathcal{I}_{A_\mathfrak{q}}$ which agree at 1 and are therefore equal. ∎

**Proposition 3.32.7 (Unique Factorisation)**
*Let $A$ be a Dedekind domain. There is a well-defined abelian group isomorphism*

$$\bigoplus_{\mathfrak{p}\in\mathcal{P}} \mathbb{Z} \xrightarrow{\sim} \mathcal{I}_A$$
$$(e_\mathfrak{p}) \longrightarrow \prod_{\mathfrak{p}:e_\mathfrak{p}\neq 0} \mathfrak{p}^{e_\mathfrak{p}}$$

*where $\mathcal{P}$ is the set of prime ideals of $A$. Write $\left(v_\mathfrak{p}\right)$ for the inverse, then it is given explicitly by*

$$v_\mathfrak{p}(I) = v_\mathfrak{p}\left(I_\mathfrak{p}\right)$$

*Further a fractional ideal $I$ is contained in $A$ if and only if $v_\mathfrak{p}(I) \geq 0$ for all prime ideals $\mathfrak{p}$.*

*Proof.* The homomorphisms exists by (3.32.6) and the universal property of the sum of abelian groups (...).

Let $I$ be a fractional ideal and define $v_\mathfrak{p}(I) \coloneqq v_\mathfrak{p}\left(I_\mathfrak{p}\right)$ using the notation of (3.31.7). This determines a group homomorphism $v_\mathfrak{p} : \mathcal{I}_A \to \mathbb{Z}$ by (3.30.8) b). We claim for every $I$ that $v_\mathfrak{p}(I)$ is zero for all but finitely many prime ideals $\mathfrak{p}$. As $I = (a)^{-1}\mathfrak{a}$ for some ideal $J$ of $A$ it is sufficient by multiplicativity to demonstrate this for ideals $\mathfrak{a}$. By (...) if $\mathfrak{a} \not\subset \mathfrak{p}$ then $\mathfrak{a}_\mathfrak{p} = A_\mathfrak{p}$ which has valuation 0. Therefore we may assume that $\mathfrak{a} \subset \mathfrak{p}$. However by (3.22.6) there are only finitely many such primes, since every non-zero prime containing $\mathfrak{a}$ must be minimal over (0), and a-fortiori over $\mathfrak{a}$.

Observe by (3.32.6) that for distinct prime ideals $\mathfrak{p}_1, ..., \mathfrak{p}_n$ we have the identity

$$\left(\mathfrak{p}_1^{r_1}...\mathfrak{p}_n^{r_n}\right)_\mathfrak{q} = \begin{cases} \left(\mathfrak{p}_k A_\mathfrak{q}\right)^{r_k} & \mathfrak{q} = \mathfrak{p}_k \\ A_\mathfrak{q} & \mathfrak{q} \neq \mathfrak{p}_k \text{ for all } k = 1...n \end{cases}$$

By the first claim there is a well-defined homomorphism $\mathcal{I}_A \to \bigoplus_{\mathfrak{p}\in\mathcal{P}} \mathbb{Z}$. We claim that this is a left inverse for the given map, that is to say

$$\prod_{\mathfrak{p}:v_\mathfrak{p}(I)\neq 0} \mathfrak{p}^{v_\mathfrak{p}(I)} = I$$

Denote the left hand side by $I'$. When $v_\mathfrak{p}(I) \neq 0$ then by the identity above we have, using (3.31.7) for the final equality,

$$I'_\mathfrak{p} = \left(\mathfrak{p}A_\mathfrak{p}\right)^{v_\mathfrak{p}(I)} = \left(\mathfrak{p}A_\mathfrak{p}\right)^{v_\mathfrak{p}\left(I_\mathfrak{p}\right)} = I_\mathfrak{p}$$

Similarly if $v_\mathfrak{p}(I) = 0$ by (3.31.7)

$$I'_\mathfrak{p} = A_\mathfrak{p} = I_\mathfrak{p}$$

We conclude by (3.30.9) that $I = I'$ as required. Finally the relation above also shows that this is a right inverse.

For the final statement by (3.30.9) we have $I \subset A \iff I_\mathfrak{p} \subset A_\mathfrak{p}$ for all prime ideals $\mathfrak{p}$, then we may use the equivalent statement over DVRs (3.31.7). ∎

**Proposition 3.32.8**
*Let $A$ be a Dedekind with $I, J$ fractional ideals. Then the following are equivalent*

*a) $I \subset J$*
*b) $J$ divides $I$ (that is there exists an ideal $\mathfrak{a}$ such that $\mathfrak{a}J = I$)*
*c) $v_\mathfrak{p}(I) \geq v_\mathfrak{p}(J)$ for all prime ideals $\mathfrak{p}$*

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*Proof.* a) $\iff$ c) By (3.30.9) we have $I \subset J \iff I_\mathfrak{p} \subset J_\mathfrak{p}$ for all prime ideals $\mathfrak{p}$. Then we may use the corresponding statement for DVRs (3.31.8).

b) $\implies$ c) Follows because $v_\mathfrak{p}(\mathfrak{a}) \geq 0$ (3.32.7).

c) $\implies$ b) We may define $\mathfrak{a} = \prod \mathfrak{p}^{v_\mathfrak{p}(I) - v_\mathfrak{p}(J)}$ which is contained in $A$ and by multiplicativity $v_\mathfrak{p}(\mathfrak{a}J) = v_\mathfrak{p}(I)$ for all prime ideals $\mathfrak{p}$ whence $\mathfrak{a}J = I$ by the correspondence in (3.32.7). ∎

**Proposition 3.32.9 (Integral Closure is a Dedekind Domain)**
*Let $L/K$ be a finite separable extension and $A \subset K$ a subring which is a Dedekind domain with quotient field $K$. Let $B$ be the integral closure of $A$ in $L$, then $B$ is a finite $A$-module and a Dedekind domain.*

*If in addition $A$ is a principal ideal domain (e.g. a discrete valuation ring) then $B$ is a free $A$-module of rank $[L : K]$.*

*Moreover $B \otimes_A K \cong L$, via $b \otimes x \mapsto bx$ (regarding $B, K \subseteq L$).*

*Proof.* From (3.24.24) we know $B$ is integrally closed and $\dim B = 1$. Then the rest follows from (3.24.26).

**Base change to $K$.** Writing $S := A \setminus \{0\}$, (3.6.31) gives an isomorphism $B \otimes_A K \cong S^{-1}B$, $b \otimes x \mapsto bx$, where $S^{-1}B$ is naturally a subring of $L$ (as $S \subset B \setminus \{0\}$ too). By (3.24.15) (applied to the integral map $A \hookrightarrow B$, and $S$), $K = S^{-1}A \to S^{-1}B$ is integral; as $S^{-1}B$ is an integral domain (a subring of $L$) and $K$ is a field, (3.24.14) gives that $S^{-1}B$ is itself a field. Being a field containing $B$, $S^{-1}B$ contains $\mathrm{Frac}(B) = L$; but $S^{-1}B \subseteq L$ already; so $S^{-1}B = L$. ∎

**Definition 3.32.10 (AKLB Situation)**
*Let $L/K$ be a finite field extension, $A \subset K$, $B \subset L$ Dedekind domains for which $A \subset B$ is integral (equivalently $B$ is the integral closure of $A$ in $L$), $\mathrm{Frac}(A) = K$ and $\mathrm{Frac}(B) = L$. We refer to this situation as "AKLB".*

**Lemma 3.32.11**
*Using the notation in (3.32.10). Let $\mathfrak{p}$ be a prime ideal of $A$, and $\mathfrak{b}$ a proper ideal of $B$. Then the following are equivalent*

*a)* $\mathfrak{p} \subset \mathfrak{b}$
*b)* $\mathfrak{b}$ *divides* $\mathfrak{p}B$
*c)* $\mathfrak{b} \cap A = \mathfrak{p}$

*and in this case we say that $\mathfrak{b}$ divides $\mathfrak{p}$ and write $\mathfrak{b} \mid \mathfrak{p}$.*

*In this case $B/\mathfrak{b}$ is naturally an $A/\mathfrak{p}$-module via the commutative diagram*

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \downarrow & & \downarrow \\ A/\mathfrak{p} & \hookrightarrow & B/\mathfrak{b} \end{array}$$

*Proof.* a) $\iff$ b) follows from (3.32.8)

a) $\implies$ c) Trivially $\mathfrak{p} \subset \mathfrak{b} \cap A$. As $1 \in \mathfrak{b}$ then $\mathfrak{b} \cap A$ is a proper ideal, and therefore equal to $\mathfrak{p}$ by maximality.

c) $\implies$ a) Immediate. ∎

**Proposition 3.32.12**
*Using the notation in (3.32.10). Let $\mathfrak{p}$ be a prime ideal of $A$. There exists at least one prime ideal $\mathfrak{q} \mid \mathfrak{p}$, and only finitely many. For every such prime ideal define the integers $e_{\mathfrak{q}/\mathfrak{p}} := v_\mathfrak{q}(\mathfrak{p}B)$, then*

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}} = \bigcap_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

*Proof.* The existence of a prime $\mathfrak{q}$ follows from the Lying Over property of integral extensions (3.24.17). The factorisation exists by (3.32.7) which also shows that there are only finitely many divisors of $\mathfrak{p}$, and the second equality holds by the Chinese Remainder Theorem (3.3.62). ∎

**Definition 3.32.13 (Ramification Index / Inertial Degree)**
*Using the notation in (3.32.10), suppose that $\mathfrak{q} \mid \mathfrak{p}$. Then we define the **ramification index** by*

$$e_{\mathfrak{q}/\mathfrak{p}} = v_\mathfrak{q}(\mathfrak{p}B)$$

*which by (...) is a positive integer. Similarly define the **inertial degree***

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$$f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{q} : A/\mathfrak{p}]$$

*which we presently show to be finite.*

**Lemma 3.32.14**
*Let $L/K$ be a finite field extension, $A$ a subring with quotient field $K$ and $B \supset A$ an overring with quotient field $L$. Suppose that $\mathfrak{p}$ is a prime ideal of $A$, $\mathfrak{b}$ is a proper ideal of $B$ containing $\mathfrak{p}$ and $A_{\mathfrak{p}}$ is a UFD. Then canonically $B/\mathfrak{b}$ is an $A/\mathfrak{p}$-module and*

$$[B/\mathfrak{b} : A/\mathfrak{p}] \leq [L : K]$$

*Furthermore we have $\mathfrak{b}B_{\mathfrak{p}} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$ and*

$$[B/\mathfrak{b} : A/\mathfrak{p}] = \left[B_{\mathfrak{p}}/\mathfrak{b}B_{\mathfrak{p}} : A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}\right]$$

*Proof.* Suppose that $\overline{x_1}, ..., \overline{x_n} \in B/\mathfrak{b}$ are linearly independent over $A/\mathfrak{p}$. It is sufficient by (...) to show that $x_1, ..., x_n$ are linearly independent over $K$. Suppose we have a linear dependence

$$0 = \sum_{i=1}^{n} \lambda_i x_i \quad \lambda_i \in K$$

Then by clearing denominators we may assume that $\lambda_i \in A_{\mathfrak{p}}$ not all zero. By removing the greatest common divisor of the non-zero elements we may assume that $(\lambda_1, ..., \lambda_n) = A_{\mathfrak{p}}$, which means at least one is not in $\mathfrak{p}A_{\mathfrak{p}}$. There is by (3.6.26) a canonical reduction map

$$\overline{\cdot} : A_{\mathfrak{p}} \to A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \xrightarrow{\sim} A/\mathfrak{p}$$

In otherwords $\overline{\lambda_i} \neq 0$, which gives a linear dependence on $\overline{x_1}, ..., \overline{x_n}$ a contradiction and we are done.

For the last statement consider the commutative diagram

$$\begin{array}{ccc} A/\mathfrak{p} & \hookrightarrow & B/\mathfrak{b} \\ \downarrow \sim & & \downarrow \sim \\ A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} & \hookrightarrow & B_{\mathfrak{p}}/\mathfrak{b}B_{\mathfrak{p}} \end{array}$$

For the vertical arrows we show they are isomorphism by (3.6.26). Observe that for $s \in A \setminus \mathfrak{p}$ we have by maximality $As + \mathfrak{p} = A$, whence $Bs + \mathfrak{p}B = B \implies Bs + \mathfrak{b} = A$. Then the last two statement follow immediately. ∎

**Lemma 3.32.15 (Local Form of Ramification Index / Inertial Degree)**
*Using the notation in (3.32.10), suppose $\mathfrak{p}$ is a prime ideal and define $A' := A_{\mathfrak{p}}$, $\mathfrak{p}' := \mathfrak{p}'A_{\mathfrak{p}}$, $B' := B_{\mathfrak{p}}$. Then $A', B'$ is also "AKLB" and there is a bijection*

$$\{\mathfrak{q} \lhd B : \mathfrak{q} \mid \mathfrak{p}\} \longleftrightarrow \{\mathfrak{q}' \lhd B'\}$$

*under which the ramification index and inertial degrees agree, i.e.*

$$e_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}'/\mathfrak{p}'}$$

*and*

$$f_{\mathfrak{q}/\mathfrak{p}} = f_{\mathfrak{q}'/\mathfrak{p}'} \leq [L : K]$$

*Further the ramification index is equal to*

$$v_{\mathfrak{q}'}(\pi B')$$

*where $\pi$ is any uniformiser for the DVR $A'$.*

*Proof.* Evidently $A'$ has quotient field $K$ and $B'$ has quotient field $L$. By definition $A'$ is a DVR and therefore a Dedekind domain. By (3.24.9) $B'$ is the integral closure of $A'$, and therefore integrally closed. It has Krull Dimension 1 by (3.24.22) and, $B$ being Noetherian (Dedekind, (3.32.1)), $B' = B_{\mathfrak{p}}$ is Noetherian too (3.15.7). Therefore we have the same situation as (3.32.10).

First consider the commutative diagram

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$$\begin{array}{ccc} A & \hookrightarrow & B \\ \downarrow & & \downarrow \\ A' & \hookrightarrow & B' \end{array}$$

regarded as subrings of $K$ and $L$ respectively.

Then by chasing the diagram $\mathfrak{p}'B' = (\mathfrak{p}B)B'$. By (3.32.12) we have a unique factorisation

$$\mathfrak{p}B = \prod_{i=1}^{n} \mathfrak{q}_i^{e_i}$$

for prime ideals $\mathfrak{q}_i$ of $B$ dividing $\mathfrak{p}$. Define $\mathfrak{q}_i' := \mathfrak{q}_i B_{\mathfrak{p}}$ then these are prime by (3.6.18) (and the fact $\mathfrak{q} \cap (A \setminus \mathfrak{p}) = \varnothing$) and contain $\mathfrak{p}'B'$ (3.32.12). Then by (3.3.50)

$$\mathfrak{p}'B' = \prod_{i=1}^{n} (\mathfrak{q}_i')^{e_i}$$

The bijection of prime ideals is well-defined by (3.6.19).

By uniqueness $e_{\mathfrak{q}'/\mathfrak{p}'} = e_{\mathfrak{q}/\mathfrak{p}}$ for all primes $\mathfrak{q} \mid \mathfrak{p}$.

The relation $f_{\mathfrak{q}'/\mathfrak{p}'} = f_{\mathfrak{q}/\mathfrak{p}} \leq [L : K]$ follows directly from (3.32.14). ∎

**Proposition 3.32.16 (Ramification Formula)**
*Using the notation of (3.32.10). For every prime ideal $\mathfrak{p}$ of $A$ we have the identity*

$$\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{p}B : A/\mathfrak{p}] \leq [L : K]$$

*Suppose in addition that $B$ is a finite $A$-module then we have equality. In particular this holds when $L/K$ is separable.*

*Proof.* Recall from (3.32.12) we have the factorisation

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

By the Chinese Remainder Theorem (3.3.62) we have a canonical isomorphism

$$B/\mathfrak{p}B \xrightarrow{\sim} \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}}}$$

Both $B/\mathfrak{p}B$ and $B/\mathfrak{q}^{e_{\mathfrak{q}}}$ are $A/\mathfrak{p}$-modules, and so equating vector space dimensions we have

$$[B/\mathfrak{p}B : A/\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} [B/\mathfrak{q}^{e_{\mathfrak{q}}} : A/\mathfrak{p}]$$

As $\mathfrak{p}B \subset \mathfrak{q}^e$ we have a decreasing sequence of $A/\mathfrak{p}$-submodules

$$B/\mathfrak{q}^e \supseteq \mathfrak{q}^1/\mathfrak{q}^e \supseteq \ldots \supseteq \mathfrak{q}^e/\mathfrak{q}^e = (0)$$

so by (3.3.130) we have

$$[B/\mathfrak{q}^e : A/\mathfrak{p}] = \sum_{i=1}^{e} [(\mathfrak{q}^{i-1}/\mathfrak{q}^e)/(\mathfrak{q}^i/\mathfrak{q}^e) : A/\mathfrak{p}]$$

We claim there is an $A/\mathfrak{p}$-module isomorphism

$$(\mathfrak{q}^{i-1}/\mathfrak{q}^e)/(\mathfrak{q}^i/\mathfrak{q}^e) \to \mathfrak{q}^{i-1}/\mathfrak{q}^i$$

This exists as an $A$-module isomorphism by (3.3.105), and therefore as an $A/\mathfrak{p}$-module isomorphism as the action factors through the quotient map $A \to A/\mathfrak{p}$. Therefore by (3.19.15)

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$$\begin{aligned}[B/\mathfrak{q}^e : A/\mathfrak{p}] &= \sum_{i=1}^{e} [\mathfrak{q}^{i-1}/\mathfrak{q}^i : A/\mathfrak{p}] \\ &= \sum_{i=1}^{e} [B/\mathfrak{q} : A/\mathfrak{p}][\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}] \\ &= \sum_{i=1}^{e} f_{\mathfrak{q}/\mathfrak{p}}[\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}]\end{aligned}$$

We claim that $[\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}] = 1$. Any non-trivial subspace of $\mathfrak{q}^{i-1}/\mathfrak{q}^i$ is a-fortiori a $B$-submodule, and by the isomorphism theorem yields an ideal between $\mathfrak{q}^{i-1}$ and $\mathfrak{q}^i$. Therefore it is enough to show there is no such ideal. By (3.6.42) we may localize at $\mathfrak{q}$ to consider the ring $B_{\mathfrak{q}}$. By definition this is a DVR and the result follows from (3.31.2). Therefore

$$[B/\mathfrak{p}B : A/\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}.$$

The inequality follows from (3.32.14).

It remains to show this equals $[B/\mathfrak{p}B : A/\mathfrak{p}] = [L : K]$ in the case $B$ is a finite $A$-module. By (3.32.15) and (3.32.14) we may reduce to the case of $A$ a discrete valuation ring, and in a particular principal ideal domain. Then by (3.24.25) we have $B$ is a free $A$-module of rank $n = [L : K]$. Let $x_1, ..., x_n$ be such a basis, then we claim that $\overline{x_1}, ..., \overline{x_1}$ is linearly independent over $A/\mathfrak{p}$ and therefore a basis for $B/\mathfrak{p}B$. For suppose

$$0 = \sum_{i=1}^{n} \overline{\lambda_i x_i} \quad \lambda_i \in A$$

then we may show that $\mathfrak{p}B = \mathfrak{p}x_1 + ... + \mathfrak{p}x_n$ whence by definition

$$\sum_{i=1}^{n} \lambda_i x_i = \sum_{i=1}^{n} \mu_i x_i$$

for some $\mu_i \in \mathfrak{p}$, from which it follows $\lambda_i = \mu_i$ and $\overline{\lambda_i} = 0$, and we are done.

The case $L/K$ is separable is then (3.24.26). ∎

## 3.32.1 Local Monogenicity

**Lemma 3.32.17 (A Uniformiser Generates a Totally Ramified Extension)**
*Using the notation of (3.32.10), with $L/K$ separable, let $\mathfrak{p}$ be a prime of $A$ with a **unique** prime $\mathfrak{q}$ of $B$ over it, of residue degree $f_{\mathfrak{q}/\mathfrak{p}} = 1$ (so $B_{\mathfrak{p}} = B_{\mathfrak{q}}$ is itself a DVR, totally ramified: $e_{\mathfrak{q}/\mathfrak{p}} = n := [L : K]$). Then for any uniformiser $\pi$ of $B_{\mathfrak{p}}$,*

$$B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi],$$

*and $\pi$'s minimal polynomial $F \in A_{\mathfrak{p}}[Y]$ over $K$ has degree exactly $n$.*

*Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.*

For $0 \leq j < n$, $\mathfrak{q}^j/\mathfrak{q}^{j+1}$ is 1-dimensional over $B_{\mathfrak{p}}/\mathfrak{q} = k$ (as $B_{\mathfrak{p}}$ is a DVR), generated by the image of $\pi^j$; residue degree 1 means $A_{\mathfrak{p}}/\mathfrak{p} = k$ too, so by downward induction on $j$, $\{1, \pi, ..., \pi^{n-1}\}$ spans $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} = B_{\mathfrak{p}}/\mathfrak{q}^n$ (as $\mathfrak{p}B_{\mathfrak{p}} = \mathfrak{q}^n$, totally ramified) as an $A_{\mathfrak{p}}/\mathfrak{p}$-module. By (3.21.6), applied to the finite $A_{\mathfrak{p}}$-module $B_{\mathfrak{p}}$, a spanning set of $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ already spans $B_{\mathfrak{p}}$ itself, so

$$B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi].$$

Let $F \in A_{\mathfrak{p}}[Y]$ be $\pi$'s minimal polynomial over $K$, of degree $d$. As $B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi]$ and $F(\pi) = 0$, $B_{\mathfrak{p}}$ is already spanned by $1, \pi, ..., \pi^{d-1}$ over $A_{\mathfrak{p}}$, so $\operatorname{rank}_{A_{\mathfrak{p}}} B_{\mathfrak{p}} \leq d$; as $B_{\mathfrak{p}}$ is $A_{\mathfrak{p}}$-free of rank $n$ (3.24.25), $d \geq n$, while $d = [K(\pi) : K] \leq [L : K] = n$ as $K(\pi) \subseteq L$. So $d = n$ exactly. ∎

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